Article

Semantic Interpretability in Deep Knowledge Tracing via Representational Grounding: Toward Individualized Learning Trajectories

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Abstract

Knowledge Tracing, which enables the estimation of how students' knowledge evolves as they interact with educational content, is a cornerstone of intelligent tutoring systems. Traditional approaches coexist with deep knowledge tracing models that deliver better predictions but suffer from a critical drawback: lack of interpretability, which is particularly problematic in educational contexts. To overcome this limitation, we propose gTransformer, a novel model that unifies the high predictive performance of deep learning with the intrinsic interpretability of traditional approaches like Bayesian Knowledge Tracing. Our design employs an encoder-decoder Transformer that enriches input sequences with parameter estimates from the interpretable model. Attention mechanisms integrate these embeddings into a latent context vector, which is projected into updated parameter values constrained to remain semantically grounded to educational constructs while incorporating rich temporal dependencies learned by the network. These enriched parameters then drive predictions through interpretable Bayesian logic. Experiments demonstrating state-of-the-art accuracy show that our model achieves superior diagnostic granularity by identifying student-specific parameters—such as initial knowledge and learning rates—that capture individual longitudinal contexts. By anchoring deep representations to defined concepts, gTransformer offers a pedagogically interpretable alternative for data-driven personalization.

Keywords: deep knowledge tracing; interpretability; grounded transformer; bayesian knowledge tracing; educational data analysis; personalized learning

1. Introduction

Knowledge Tracing (KT) constitutes a foundational problem in Artificial Intelligence in Education (AIED) and Intelligent Tutoring Systems (ITS). Fundamentally, it defines the task of modeling the dynamic evolution of a student's knowledge state over time based on their history of interactions with learning materials [1]. By estimating the probability that a student has mastered a specific skill at a given moment, KT systems enable the delivery of personalized instruction—recommending remedial content for struggling students or accelerating the curriculum for those who demonstrate proficiency. In an era where educational environments are becoming increasingly digital and diverse, the ability to accurately

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track and interpret student mastery is not merely a technical optimization but a critical requirement for effective, scalable, and learner-centred education.

Historically, the development of KT has been characterized by a dichotomy between two distinct paradigms [2]. The first, which predominated in the field for years, is characterized by probabilistic models such as Bayesian Knowledge Tracing [1] and Factor Analysis Models based on logistic regression—including the Additive Factor Model [3], Performance Factor Analysis (PFA) [4], and Knowledge Tracing Machine (KTM) [5]. Factor Analysis Models are theoretically supported by the Item Response Theory (IRT) [6], which has played a large role in educational assessment and measurement. These traditional approaches are intrinsically interpretable: they model learning as a process governed by explicit parameters related to pedagogical, cognitive or psychometric principles. Because the model structure mirrors a theoretical understanding of human learning, educators can inspect these values to validate the model's reasoning and trust its outputs. However, this transparency comes at the cost of representational rigidity; the simplifying assumptions required by these models often limit their predictive power, causing them to struggle with the complex, non-linear, and long-range dependencies inherent in real-world student behavior [7].

The second paradigm emerged with the advent of Deep Knowledge Tracing (DKT) [8], which shifted the focus to high-dimensional models. By leveraging Recurrent Neural Networks first and, more recently, Transformer architectures [9], DKT models treat student interaction histories as complex temporal sequences. These models have achieved state-of-the-art predictive performance, significantly outperforming classical baselines by automatically extracting latent features from vast amounts of data [2]. However, this substantial gain in predictive performance has frequently come at the cost of model transparency, creating a significant interpretability gap. Deep learning models are notoriously opaque "black boxes", where the internal high-dimensional representations bear no direct correspondence to constructs that are interpretable in terms of human-understandable concepts or established domain knowledge. In a high-stakes domain such as education—where algorithmic decisions influence learning trajectories, grades, and opportunities—this lack of transparency poses critical challenges for practical adoption.

In the following section, we introduce gTransformer (*grounded Transformer*), a novel architecture designed to satisfy the requirements of Semantic Interpretability and propose a *Representational Grounding* methodology to align iDKT estimations with educational constructs defined by a reference theory. To validate the approach, we define a rigorous metric for interpretability and analyze the trade-off between predictive performance and interpretability via a systematic *Pareto frontier* sweep. Our results demonstrate that iDKT achieves high theoretical alignment while preserving over 99% of the predictive capacity of unconstrained baselines. Notably, on the ASSISTments 2009 dataset, our experiments revealed that enhancing interpretability resulted in a performance gain, with an observed increase of +0.012 in AUC. These findings provide robust empirical support for the assertion by Rudin [10] that the inherent trade-off between predictive accuracy and model interpretability is often a myth, particularly in domains with significant underlying structure. This perspective is further supported by the *Rashomon Set* argument [11], which posits that for many real-world problems, the set of high-performing models is sufficiently large to contain solutions that are both accurate and intrinsically interpretable.

The validation of our approach will be guided by three research hypotheses:

- **H1 (Representational Grounding)**: Through Representational Grounding, the internal representations of an attention-based Transformer model can be aligned with pedagogically meaningful constructs to achieve Semantic Interpretability. We hypothesize that iDKT can achieve this kind of interpretability by exhibiting high linear recoverability with respect to BKT parameters across diverse educational datasets. Specifically, we posit that BKT constructs can be accurately recovered from iDKT's representations using simple linear transformations, which we validate through diagnostic probing with control tasks.

- **H2 (Predictive Performance-Interpretability Trade-off)**: The pursuit of intrinsic interpretability does not necessarily imply a significant degradation in predictive performance. We can identify configurations that preserve near-optimal accuracy while achieving high semantic alignment. Furthermore, we posit that in certain contexts, the inductive bias introduced by pedagogical grounding can act as a regularizer, actually enhancing the model's predictive accuracy.
- **H3 (Estimations Granularity)**: Besides the alignment with a reference model, Representational Grounding enables the discovery of fine-grained parameter values. Specifically, we hypothesize that iDKT provides superior diagnostic granularity by inferring personalized learning rates and initial mastery levels that capture behavioral heterogeneity more effectively than population-level probabilistic models.

The remainder of this paper is structured as follows. Section 3 describes the iDKT model, detailing the representational grounding mechanism and the multi-objective loss system. Section 4 presents the experimental setup. Section 5 discusses the empirical results, and Section 6 presents our final conclusions.

2. Related Work

2.1. Bayesian Knowledge Tracing

For this study, we selected Bayesian Knowledge Tracing (BKT) as our anchoring theoretical framework. BKT models the learning process as a hidden Markov process governed by four explicit parameters: *Initial Mastery*, the probability a student knows a skill prior to practice; *Learn Rate*, the probability of transitioning from unlearned to learned states after a practice opportunity; *Guess*, the probability of a correct response despite a lack of mastery; and *Slip*, the probability of an incorrect response despite mastery. These parameters provide a causal narrative of student performance that is intuitive to educators and aligned with cognitive science principles.

Standard BKT suffers from a significant limitation: it typically operates at a population level, estimating a single set of parameters for all students interacting with a given skill. While *Individualized BKT* approaches exist, they are often computationally expensive and struggle with data sparsity [12]. The Representational Grounding used in the iDKT model offers a novel way to address individualization by leveraging the capabilities of Transformers to compress student interaction histories into a context vector. This allows the model to infer personalized initial knowledge and learning rates dynamically, transforming population-level BKT insights into high-granularity student diagnostics.

2.2. Interpretability in Deep Knowledge Tracing

Interpretability in DKT remains an open issue, despite diverse attempts categorized by Bai et al. [13] into two primary modalities: *post-hoc* and *ante-hoc* approaches. Post-hoc methods attempt to explain a trained black-box model using techniques such as Layer-wise Relevance Propagation (LRP) [], perturbation analysis [], or local approximations like LIME []. However, as argued by Rudin [10], relying on post-hoc explanations for high-stakes decision-making is fundamentally flawed. These methods have limited utility for teachers and domain experts since they produce explanations related to the model's internal structures that are disconnected from pedagogically sound principles, thus requiring significant technical expertise to interpret.

The second category, *ante-hoc* methods, aims for intrinsic interpretability, encompassing two main approaches: traditional models that rely on simple, transparent structures, and DKT variants that achieve interpretability by incorporating additional modules or side information. Prominent examples in the latter group include Self-Attentive Knowledge Tracing (SAKT) [14], the first model to incorporate attention mechanisms, and Attentive Knowledge Tracing (AKT) [15], which achieved high predictive performance by integrating attention heads that apply different temporal distances to capture student interaction patterns with varying temporal resolution. However, it can be maintained that relying solely on attention mechanisms presents similar practical obstacles to post-hoc methods since they do not provide a clear interpretation of the model's predictions in terms of pedagogically sound principles.

Concerning the integration of side information into ante-hoc methods, Item Response Theory (IRT) [6] represents the most prominent theoretical foundation. IRT estimates the probability of a student correctly answering a question given their latent ability and the question's difficulty. This framework has been coupled with DKT models such as Deep-IRT [16], which leverages dynamic key-value memory networks to capture learners' trajectories; this allows for the inference of abilities and item difficulties via neural networks, which are subsequently operationalized to predict answer correctness. Similarly, TC-MIRT [17] embeds the parameters of a multidimensional IRT to predict students' states and generate interpretable parameters. Other approaches, such as KIKT [18], harness IRT to simulate learners' performance, while LANA [19] adopts a IRT variant to cluster students' ability levels. Finally, QIKT [20] features question-centric representations combined with an interpretable IRT layer. Beyond IRT, other researchers have leveraged diverse principles, such as constructive learning [21,22], learning curve and forgetting curve [23], finite state automaton (FSA) [24], classical test theory (CTT) [25], and monotonicity theory [26].

2.3. Current Limitations

While providing insights into controlled scenarios, these approaches may struggle to encapsulate the multifaceted and dynamic nature of learning processes and the efficacy is often constrained by the specificity and scope of the underlying theories [13]. Moreover, the mere incorporation of principles from psychology or pedagogy does not guarantee interpretability. As noted by Von Rueden et al. [27], improved interpretability remains a secondary objective in most knowledge integration approaches used in machine learning, often overshadowed by performance optimization.

Informed Machine Learning

We present gTransformer, a deep learning model based on an encoder-decoder Transformer architecture [9] extended with specialized components designed to achieve Semantic Interpretability. Within the framework of explainability techniques for Knowledge Tracing proposed by Bai et al. [13], our proposal is an ante-hoc model that provides model-intrinsic interpretability through the integration of attention mechanisms. However, unlike existing methods in this category, primarily intended to deep learning specialists, gTransformer is specifically oriented toward domain experts by providing estimations grounded in educational theory.

The integration of domain knowledge into deep learning has been broadly explored through conceptual frameworks such as theory-guided data science [28], physics-informed machine learning [29], and informed machine learning (IML) [27]. iDKT operationalizes IML principles within the field of Knowledge Tracing, aligning with the taxonomic dimensions of information flow characterized by Von Rueden et al. [27], as shown in Figure 1.

Table Our Approach vs Other

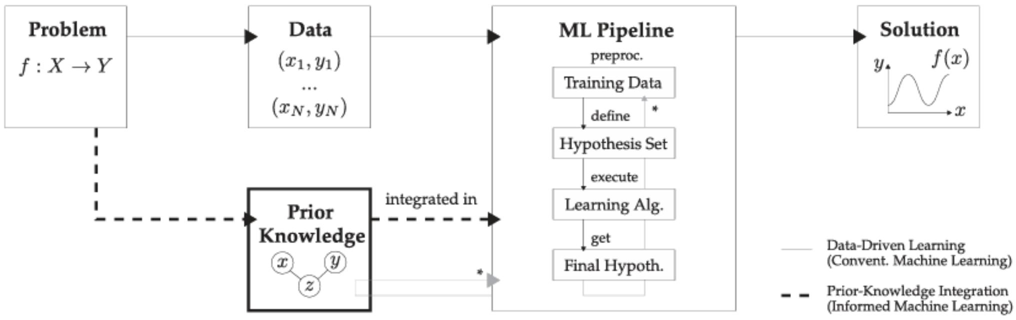


Figure 1. The information flow in iDKT, following the taxonomy proposed by Von Rueden et al. [27], comprises: (1) *Prior Knowledge*: simulation results from an educational model serve as a source of structured information; (2) *Training Data*: the augmented data is used to train the model; (3) *Hypothesis Set*: an attention-based Transformer architecture provides the inductive bias required to align internal representations with the reference model; (4) *Learning Algorithm*: optimization is driven by a compound loss function, which guides the system toward semantically aligned representations; and (5) *Final Hypothesis*: the internal representations are projected and compared with the prior knowledge of the reference model.

3. The gTransformer Model

3.1. Architecture

The core architecture of iDKT, illustrated in Figure 2, builds upon the standard Transformer framework [9] by integrating specialized modules designed to facilitate multi-objective optimization. The architecture is structured into the following functional layers:

1. *Input Data*: this stage handles the ingestion of data augmented with prior knowledge from the reference model. Student interactions (concepts, questions, and binary responses $r \in \{0, 1\}$) are enriched with BKT predictions (p_{BKT}) and the per-skill parameters detailed in section **??**, namely: *Initial Mastery* ($P(L_0)$), *Learn Rate* ($P(T)$), *Guess* ($P(G)$), and *Slip* ($P(S)$).
2. *Embeddings*: this layer constitutes the central component of iDKT, providing the foundational material for the synthesis of the attention-driven context vector. By applying the transformations detailed in Section 3.2, base embeddings ($x_{1:t-1}$) and ($y_{1:t-1}$) are augmented into individualized embeddings ($x'_{1:t-1}$) and ($y'_{1:t-1}$) to capture high-granularity patterns of student behavior. This semantic enrichment enables the interaction history to be encoded into a meaningful context vector (z_t) that encapsulates the nuances of the student's behavior throughout the sequence.
3. *Attention Heads*: the model implements an encoder-decoder Transformer architecture with various attention mechanisms. The encoder processes the individualized interaction history ($y'_{1:t-1}$) through self-attention to capture global student behavior patterns. The decoder processes the current individualized task (x'_t) by passing it through self-attention blocks (processing the question representation) and cross-attention blocks (attending to the encoded interaction history). This cross-attention mechanism synthesizes both streams into a contextualized representation.
4. *Prediction Head and Grounded Read-out*: the system generates dual diagnostic outputs. The *Prediction Head* (h_{pred}) utilizes the contextualized representation (z_t) generated by the Transformer to output the final performance probability (p_{pred}). Simultaneously, a *Grounded Read-out* provides the estimated proficiency (p_{know}) and rate (p_{rate}) parameters, which characterize the student's individualized profile by directly projecting the mean values of the individualized embeddings.

5. *Loss Functions*: optimization is driven by a multi-objective loss pipeline. While the prediction head is supervised by $\mathcal{L}_{sup}$, the diagnostic read-out is guided by $\mathcal{L}_{int}$, which penalizes deviations between the inferred factors (p_{know}, p_{rate}) and the BKT grounding values. Since these factors govern the individualized embedding layer (see Section 3.2), this mechanism constraints the entire latent space to remain semantically aligned with the reference model while preserving the Transformer's capacity for complex pattern discovery.

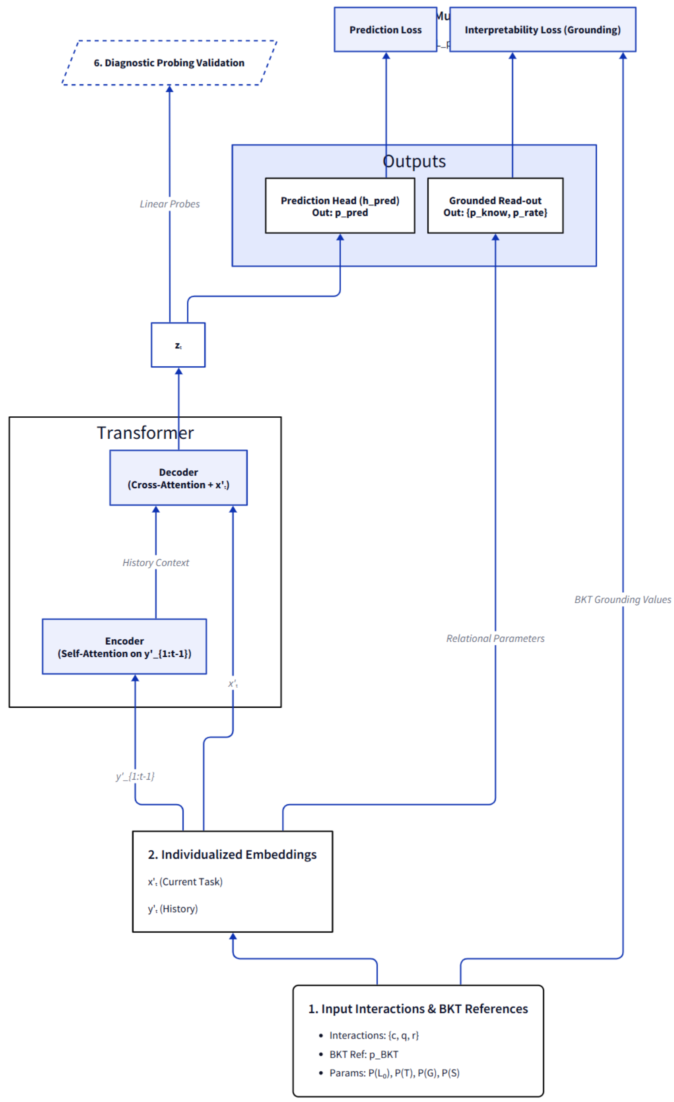


Figure 2. The diagram illustrates the functional layers of the iDKT Architecture: (1) *Input Layer*: data enriched with BKT reference data (p_{BKT} and theoretical parameters); (2) *Embeddings*: provides the foundational material through individualized transformations (x', y'); (3) *Attention Heads*: encoder-decoder transformer synthesizes history and task into a contextualized representation (z_t); (4) *Prediction Head and Grounded Read-out*: provide performance probabilities (p_{pred}) from historical context and grounded pedagogical diagnostics (p_{know}, p_{rate}) from representational grounding; (5) *Loss Functions*: implement a multi-objective pipeline ($\mathcal{L}_{sup}, \mathcal{L}_{int}$) to ensure semantic alignment; and (6) *Diagnostic Probing Validation*: verifies the structural preservation of pedagogical constructs through linear probing of internal representations.

3.2. Embeddings

In standard educational datasets, such as ASSISTments 2009, ASSISTments 2015, Algebra 2005, and others [30], student interactions are recorded at the level of specific questions or tasks, each of which is associated with one or more underlying concepts or knowledge components. This structure reflects the fact that proficiency in a concept (e.g., the Pythagorean Theorem) is acquired through interactions with a diverse range of tasks. While all tasks involving a concept share the same semantic core, they differ in their specific manifestations—most notably in their intrinsic difficulty or complexity. Therefore, an effective representation must capture both the shared identity of the concept and the unique deviation of the specific task.

In Transformer-based models we can operationalize this principle representing the tasks as embedding vectors:

$$x' = c + u \cdot d \quad \text{(Task)} \tag{1}$$

In this formulation, c acts as the *Concept Base*, a vector representing the invariant semantic identity of the concept while the vector d represents the learnable *Question Variation Axis*, defining the specific direction of the "transition gap". The scalar u serves as the *Relational Magnitude*, representing the question's specific relative difficulty compared to other questions involving the same concept. Consequently, rather than encode simple embeddings for every question, we encode the vector sum of two distinct components with clear semantic meaning: the base *concept identity* and the *difficulty shift*.

In a similar way, we can operationalize the interactions between questions and students as embedding vectors:

$$y' = e + u \cdot (f + d) \quad \text{(Interaction History)} \tag{2}$$

Here e represents the *Interaction Base*, which is a combined representation of concept c and the binary outcome r (correct/incorrect), while f represents the *Interaction Variation Axis*, which is similar to the Question Variation Axis (d_c) but is specific to the interaction between a question and a student. The inclusion of d also here ensures that the difficulty vectors are consistent across both questions (x') and interactions (y').

Extending this rationale, we can enrich the x' embeddings by integrating additional components with explicit semantic significance. Specifically, by adopting Bayesian Knowledge Tracing (BKT) as a reference model, we can incorporate vectors corresponding to its core theoretical parameters—Initial Knowledge (L_0) and Learning Rate (T)—thereby grounding the deep representation in established pedagogical constructs.

To get individualized values for these parameters, we decompose them into population-level bases and student-specific deviations:

$$l_c = L_0 + k_s \cdot d_k \quad \text{(Personalized Initial Knowledge)} \tag{3}$$

$$t_c = T + v_s \cdot d_v \quad \text{(Personalized Learning Rate)} \tag{4}$$

where l_c is the personalized initial knowledge for concept, t_c is the personalized learning rate for concept, L_0 and T are the population-level base embeddings, d_k and d_v are the learnable variation axes vectors (similar to the difficulty axis d), and k_s and v_s are the scalar student-specific deviations learned for each individual.

Finally, we include these vectors to get the individualized input embeddings for the encoder and decoder components of the Transformer:

$$x' = (c + u \cdot d) - l_c \quad \text{(Individualized Task)} \tag{5}$$

$$y' = (e + u \cdot (f + d)) + t_c \quad \text{(Individualized Interaction History)} \tag{6}$$

We can interpret these individualized embeddings as follows:

- *Transition Gap* (x'): Represents the individualized difficulty of a task by offsetting objective item difficulty with the student's proficiency (l_c). Through this relational lens, task demands are normalized against the individual's baseline, effectively isolating the residual 'gap' that the student must bridge to achieve mastery.
- *Transition Momentum* (y'): Captures individualized momentum by modulating observed outcomes with the learning rate (t_c). This allows the encoder to process interactions within the context of the student's acquisition rate, enabling the model to differentiate between students who share similar performance histories but different acquisition rates.

3.3. Loss Functions

The optimization of iDKT is governed by a multi-objective loss function designed to ensure that the high-capacity Transformer architecture remains aligned with pedagogical principles. Following established patterns in theory-guided machine learning [28,31], we incorporate theoretical constraints as grounding terms in the optimization objective. This approach steers the model to learn latent representations that are both predictive of student outcomes and semantically consistent with established educational constructs. The total loss $\mathcal{L}_{total}$ is defined as a weighted sum of supervised, grounding, and regularization components:

$$\mathcal{L}_{total} = \mathcal{L}_{sup} + \lambda_{ref}\mathcal{L}_{ref} + \sum_{k \in \{know, rate\}} \lambda_k \mathcal{L}_k + \lambda_{reg}\mathcal{L}_{reg} \tag{7}$$

3.4. Supervised Alignment ($\mathcal{L}_{sup}$)

The primary objective $\mathcal{L}_{sup}$ uses standard Binary Cross-Entropy (BCE) between the iDKT performance predictions $\hat{y}_t$ and the observed ground truth outcomes $r_t \in \{0, 1\}$. This loss ensures predictive accuracy by minimizing the deviance from observed student responses:

$$\mathcal{L}_{sup} = -\frac{1}{N}\sum_{t=1}^{N}[r_t \log(\hat{y}_t) + (1 - r_t)\log(1 - \hat{y}_t)] \tag{8}$$

3.4.1. Interpretability and Grounding Losses

To satisfy the criteria for Semantic Interpretability defined in Section **??**, iDKT utilizes a dual grounding mechanism that anchors both the model's internal representations and its external behavior to the reference BKT logic.

Representational Grounding ($\mathcal{L}_{know}$, $\mathcal{L}_{rate}$): These losses utilize Mean Squared Error (MSE) to anchor the student-level diagnostic read-outs (p_{know}, p_{rate}) directly to the reference model's theoretical parameters. This ensures that the latent shifts introduced in the grounding layer correspond to explicit pedagogical constructs:

$$\mathcal{L}_{know} = \text{MSE}(p_{know}, P(L_0)) \tag{9}$$
$$\mathcal{L}_{rate} = \text{MSE}(p_{rate}, P(T)) \tag{10}$$

where $p_{know} = \sigma(\bar{l}_c)$ and $p_{rate} = \sigma(\bar{t}_c)$ are obtained by calculating the mean across the d_{model} latent embedding dimensions followed by a sigmoid projection. This mechanism ensures that iDKT does not learn arbitrary weights but instead "internalizes" the semantic identity of the theoretical constructs.

Behavioral Alignment ($\mathcal{L}_{ref}$): In addition to internal grounding, iDKT enforces a behavioral constraint on final predictions. $\mathcal{L}_{ref}$ minimizes the deviance between the iDKT predictions ($\hat{y}$) and the referece BKT predictions (p_{BKT}):

$$\mathcal{L}_{ref} = \text{MSE}(\hat{y}, p_{BKT}) \tag{11}$$

By penalizing behavioral divergence, this alignment loss forces a *fidelity chain* through the model: the Transformer must utilize the grounded internal representations to make final predictions that are consistent with the reference logic.

3.5. Regularization Term ($\mathcal{L}_{reg}$)

While grounding losses anchor the global structure of the latent space, $\mathcal{L}_{reg}$ ensures that student-level individualization remains *parsimonious*. This loss penalties the deviation from the theoretical prior, implementing a *normal student prior* where the model assumes subjects adhere to population-level BKT parameters unless evidence from their unique interaction history justifies a deviation. We apply L_2 penalties to the scalar parameters governing these deviations:

$$\mathcal{L}_{reg} = \lambda_{\text{u}} \sum_{q \in Q} u_q^2 + \lambda_{\text{sep}}(k_s^2 + v_s^2) \tag{12}$$

where u_q represents item difficulty variance, and $\{k_s, v_s\}$ are the student-specific deviations for proficiency and rate. This constraint prevents overfitting to noise in sparse student trajectories.

4. Experimental Setup

The experimental validation of iDKT is guided by the following research questions:

4.1. Implementation Details

We implemented the iDKT model in `PyTorch` and used the benchmark library `PYKT` [30] to leverage standardized data preprocessing, dataset splitting and benchmarking of baseline models. For training, we used standard 5-fold cross-validation with an 80/20 train/test split. The model was trained using the Adam optimizer with a learning rate of $1e-4$, a batch size of 64, and a dropout rate of 0.2 to prevent overfitting. The maximum number of epochs was set to 200, with an early stopping mechanism (patience=10) to terminate training if validation performance plateaued.

The Transformer architecture was configured with an embedding dimension d_{model} of 256, 8 attention heads, and 4 encoder/decoder blocks. The feed-forward dimension d_{ff} was set to 512. Regularization penalties for individualization parameters (L_2 on u_q, k_s, v_s) were all set to $1e-5$. For reproducibility, all experiments were seeded (seed=42) and executed on NVIDIA A100 GPU infrastructure. Predictive performance was evaluated using Area Under the ROC Curve (AUC) and Accuracy (ACC), while interpretability was assessed using the metrics defined in Section 5.2.

4.2. Datasets

We evaluated iDKT across four widely used educational datasets following the S-protocol for sequence length consistency [30]:

- ASSISTments 2009 (AS2009): A benchmark dataset from the ASSISTments platform (2009-2010), representing a standard for knowledge tracing evaluation [32].
- ASSISTments 2015 (AS2015): A large-scale release from the same platform with a high volume of student records [32].
- Bridge to Algebra 2006 (Bridge2006): A dataset from the KDD Cup 2010 EDM Challenge focused on the Carnegie Learning Bridge to Algebra system [33].
- NeurIPS 2020 Education Challenge (NIPS34): A diagnostic dataset from the Eedi platform containing multiple-choice math questions [34].

5. Results and Discussion

5.1. Predictive Performance

To evaluate the predictive performance of the iDKT model, we used the Area Under the Curve (AUC) and the Classification Accuracy (ACC) of the models on the test set with the 5 datasets described in Section 4.2. The results are shown in the Table 1.

Table 1. Predictive Performance across the evaluated datasets (Mean AUC and Mean ACC).

Model	AS2009		AS2015		Bridge2006		NIPS34	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
iDKT	0.8240	0.7573	0.7224	0.7520	0.7879	**0.7565**	0.7501	0.6868
AKT	0.8271	**0.7620**	0.7246	0.7540	**0.7934**	0.6947	**0.7745**	**0.7081**
ATKT	**0.8286**	0.7579	**0.7755**	**0.7637**	0.6911	0.6395	0.7391	0.6783
DKT	0.8208	0.7547	0.7268	0.7536	0.6701	0.6209	0.7354	0.6772
DKT+	0.8216	0.7570	0.7284	0.7544	0.6693	0.6121	0.7362	0.6781
DKVMN	0.8082	0.7476	0.7174	0.7535	0.6741	0.6522	0.7208	0.6646
SAINT	0.7638	0.7207	0.7031	0.7483	0.7340	0.7152	0.7127	0.6563
SAKT	0.6955	0.6608	0.6810	0.7405	0.5627	0.6328	0.6686	0.6234

When comparing these results with baseline architectures, we observe that iDKT achieves highly competitive performance across all benchmarks. In terms of Mean AUC, iDKT consistently ranks among the top performers, securing the 2nd position on the Bridge2006 and NIPS34 datasets—surpassed only by AKT model—and the 3rd position on AS2009. Notably, in terms of Classification Accuracy (ACC), iDKT outranks all baseline models on the Bridge2006 dataset, achieving the highest value of 0.7565. These results provide empirical evidence that the models' pedagogical grounding does not compromise its predictive integrity, maintaining state-of-the-art performance while providing additional layers of interpretability. Furthermore, to test the hypothesis that interpretability constraints might degrade accuracy, we evaluated an *unconstrained* variant of iDKT (with all grounding loss terms $\lambda_{\text{ref}}, \lambda_{\text{init}}, \lambda_{\text{rate}}$ set to zero). On the primary AS2009 dataset, this unconstrained baseline achieved a Mean AUC of 0.8065, significantly lower than the grounded model's 0.8240. This result suggests that the pedagogical regularization acts as a beneficial inductive bias, guiding the optimization toward more generalizable minima rather than hindering performance.

5.2. Interpretability Validation (RQ1)

To verify that iDKT's internal representations—specifically Personalized Initial Knowledge ($\mathbf{l}_c$) and Personalized Learning Rate ($\mathbf{t}_c$) as defined in Section 3.2—faithfully represent the educational constructs postulated by the reference model, we evaluate the representational alignment through three metrics that quantify the preservation and functional utility of pedagogical constructs.

1. *Convergent Validity (Representational Alignment)*: This metric, denoted as $\mathcal{I}_1$, measures the statistical correspondence between iDKT's latent projections and the reference model's theoretical parameters. We calculate the Pearson correlation between the

student-level estimations and the BKT-derived priors for both initial mastery and learning rate. The metrics are expressed as:

$$\mathcal{I}_{1l} = \text{Corr}(p_{know}, P(L_0)) \quad \text{and} \quad \mathcal{I}_{1t} = \text{Corr}(p_{rate}, P(T)) \tag{13}$$

where $p_{know} = \sigma(\bar{l}_c)$ and $p_{rate} = \sigma(\bar{t}_c)$ are the mean values across the latent embedding dimensions projected through a sigmoid activation.

2. *Predictor Equivalence (Functional Substitutability)*: This metric, denoted as $\mathcal{I}_2$, assesses whether the learned iDKT parameters can functionally substitute the reference parameters in a simulation. We "inject" the projected factors (p_{know} and p_{rate}) into the canonical BKT recurrence equations to generate *induced mastery trajectories*. $\mathcal{I}_2$ is defined as the correlation between these induced trajectories ($P(L)_{ind}$) and the reference trajectories ($P(L)_{ref}$) generated by the original BKT model:

$$\mathcal{I}_2 = \text{Corr}\Big(P(L)_{ind}, P(L)_{ref}\Big) \tag{14}$$

Table 2 summarizes the alignment metrics for the unconstrained iDKT model ($\lambda_{\mathcal{I}} = 0$), demonstrating that without explicit grounding, the deep model's latent space remains decoupled from pedagogical constructs.

Table 2. Interpretability Alignment Metrics for unconstrained iDKT ($\lambda_{\mathcal{I}} = 0$).

Dataset	$\mathcal{I}_{1l}$	**Int.**	$\mathcal{I}_{1t}$	**Int.**	$\mathcal{I}_2$	**Int.**
AS2009	-0.1382	Poor	-0.0067	Negl.	0.1870	Poor
AS2015	-0.0392	Negl.	0.0908	Negl.	0.0975	Negl.
Bridge2006	-0.0645	Negl.	0.0160	Negl.	0.0809	Negl.
NIPS34	0.2070	Poor	-0.0425	Negl.	0.1722	Poor

Enabling Representational Grounding ($\lambda_{\mathcal{I}} = 0.10$) leads to a significant restoration of semantic alignment, as evidenced in Table 3.

Table 3. Interpretability Alignment Metrics for grounded iDKT ($\lambda_{\mathcal{I}} = 0.1$).

Dataset	$\mathcal{I}_{1l}$	**Int.**	$\mathcal{I}_{1t}$	**Int.**	$\mathcal{I}_2$	**Int.**
AS2009	0.5409	Good	0.3131	Fair	0.6250	Excell.
AS2015	0.9217	Excell.	0.8801	Excell.	0.1749	Poor
Bridge2006	0.4561	Fair	0.6422	Good	0.0859	Negl.
NIPS34	0.4912	Fair	0.4105	Fair	0.2104	Poor

The results presented in Tables 2 and 3 reveal a notable divergence between *Convergent Validity* ($\mathcal{I}_1$) and *Predictor Equivalence* ($\mathcal{I}_2$). Although $\mathcal{I}_1$ attains high levels across most datasets even with relatively low $\lambda_{\mathcal{I}}$ grounding weights, $\mathcal{I}_2$ remains consistently low.

High $\mathcal{I}_1$ scores demonstrate that iDKT successfully internalizes the *semantic identity* of the theoretical constructs (Initial Knowledge and Learning Rate). However, $\mathcal{I}_2$ measures functional substitutability within a comparatively constrained reference model that is unable to capture the intricate behavioral patterns identified by iDKT. As a high-capacity Transformer, iDKT captures complex, long-range dependencies and context-aware dynamics that exceed the modeling capability of classical BKT. This discrepancy provides empirical evidence that iDKT does not merely mimic the reference model, but rather maps its core pedagogical constructs onto a more sophisticated and predictive architecture.

5.2.1. Latent Structure Validation via Diagnostic Probing

While the convergent validity metrics ($\mathcal{I}_1, \mathcal{I}_2$) provide evidence that the model's inputs are successfully grounded, a question remains: does this pedagogical signal survive the

Transformer's deep processing? To verify this, we employ *Diagnostic Probing with Control Tasks* [35] to validate the *Structural Encoding* of constructs at the point of prediction.

The probing protocol investigates whether the theoretical constructs injected into the embedding layer (Equations 3 and 4) remain dominant within the final contextualized representations (z_t) used for performance prediction. If the information remains linearly recoverable from z_t, it proves that the Transformer built its predictive logic upon the grounded foundations.

To validate this, we extract the task-contextualized representations (z_t)—the concatenation of the decoder's hidden state and the task embedding—and train a simple linear regression probe to recover the BKT mastery probability $P(L_t)$. However, high performance on this *true task* alone is insufficient, as high-capacity models might memorize arbitrary patterns. To rigorously distinguish genuine encoding from superficial correlation, we implement a *control task* [35] by training an identical probe to predict a randomly shuffled version of the theoretical labels. We quantify the integrity of the representation via the *Selectivity* metric:

$$\text{Selectivity} = R^2_{\text{true}} - R^2_{\text{control}} \tag{15}$$

where R^2_{true} measures the variance explained when predicting actual BKT mastery, and R^2_{control} measures the same for the shuffled control. A high positive selectivity score ($\Delta R^2 > 0.5$) provides terminal evidence of *Structural Encoding*, confirming that the pedagogical constructs remain the primary informative features through the model's entire reasoning chain.

Table 4 presents the probing results for the grounded iDKT model ($\lambda_{\mathcal{I}} = 0.1$) on two representative datasets.

Table 4. Diagnostic Probing Results for grounded iDKT ($\lambda_{\mathcal{I}} = 0.1$).

Dataset	R^2_{true}	R^2_{control}	Selectivity (ΔR^2)	Interpretation
AS2009	0.6731	-0.0139	**0.6870**	Robust Structural Encoding
AS2015	0.6472	-0.0043	**0.6515**	Robust Structural Encoding

The results provide compelling validation of iDKT's interpretability. Both datasets exhibit selectivity scores exceeding 0.65, substantially above the threshold for "strong encoding" ($\Delta R^2 > 0.5$) established in the literature [35]. The slightly negative control task performance confirms that the model is not simply memorizing arbitrary patterns, but has genuinely organized its post-attention representations around meaningful constructs. Critically, these probing results validate that the grounded information embedded in the input layer (via l_c and t_c) is preserved through the Transformer's self-attention mechanism and remains structurally encoded in the final contextualized representations. This demonstrates that the representational grounding fundamentally shapes how the model processes and represents student knowledge states.

This dual validation—combining correlation-based semantic alignment ($\mathcal{I}_1, \mathcal{I}_2$) with rigorous probing selectivity—establishes that iDKT satisfies the requirements of Semantic Interpretability defined in Section **??**. The model's representations are both semantically meaningful (aligned with BKT constructs) and structurally validated via control tasks, providing a robust foundation for pedagogically interpretable predictions.

5.3. Accuracy–Interpretability Trade-Off (RQ2)

A common assumption in deep knowledge tracing is that predictive performance and interpretability are opposing goals. To explore this relationship, we construct a *Pareto frontier* by systematically varying the *Alignment Intensity* ($\lambda_{\mathcal{I}}$), a unified hyperparameter that scales the weights of all three interpretability loss components simultaneously: $\lambda_{ref} = \lambda_{know} = \lambda_{rate} = \lambda_{\mathcal{I}}$. By varying $\lambda_{\mathcal{I}} \in [0, 1]$, we map the resulting effects on both accuracy and alignment.

We measure the overall interpretability of the model using a *Composite Alignment* metric ($\bar{\mathcal{I}}$), defined as the arithmetic mean of the representational ($\mathcal{I}_{1l}, \mathcal{I}_{1t}$) and functional ($\mathcal{I}_2$) components:

$$\bar{\mathcal{I}} = \frac{1}{3}(\mathcal{I}_{1l} + \mathcal{I}_{1t} + \mathcal{I}_2) \tag{16}$$

Tables 5 and 6 summarize the results for the AS2009 and AS2015 datasets across the $\lambda_{\mathcal{I}}$ sweep.

Figures 3 and 4 illustrate the resulting frontiers. We observe that Representational Grounding serves as an effective regularizer, although the *Saturation Point*—the weight at which interpretability is maximized without significant loss of fidelity—depends on dataset complexity. For the problem-level AS2009 dataset, we observe a synergy point at $\lambda_{\mathcal{I}} = 0.10$ where both metrics improve. Conversely, the concept-level AS2015 dataset reaches saturation at lower constraints, retaining 99.9% of baseline AUC. Beyond these points, a *grounding collapse* starts, where excessive theoretical constraints force the model to ignore nuanced behavioral signals in favor of rigid theoretical priors.

Table 5. Pareto Sweep results for AS2015.

$\lambda_{\mathcal{I}}$	AUC	Alignment ($\bar{\mathcal{I}}$)	Interpretation
0.0 (Baseline)	0.7248	-0.0012	Black Box
0.1 (Saturation)	0.7245	0.6673	99.9% AUC Retained
0.2	0.7181	0.6453	High-Fidelity Diagnostic
0.3	0.7112	0.6095	Latent Degradation
0.5	0.6975	0.5691	Theory-Dominant
0.8	0.6830	0.4828	Grounding Collapse
1.0	0.6763	0.4828	Grounding Collapse

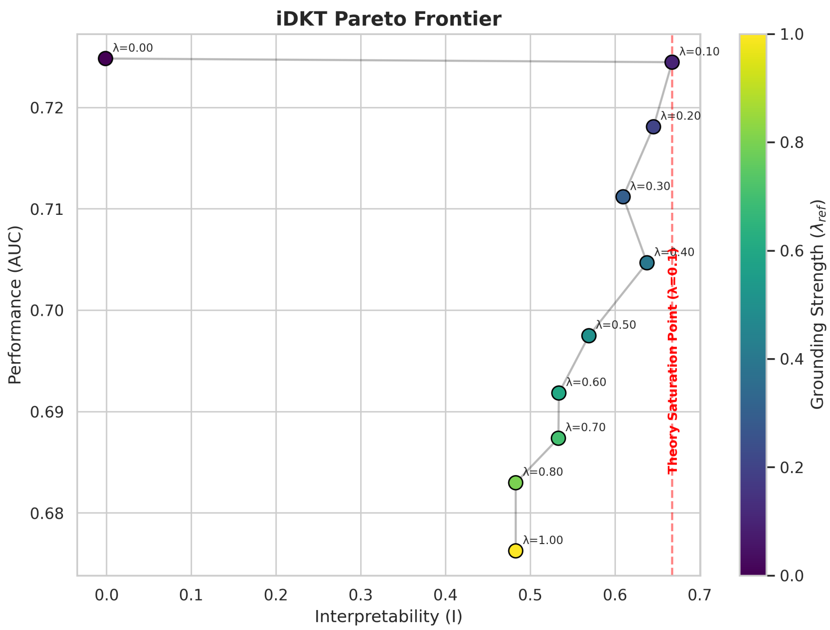


Figure 3. iDKT Pareto Frontier for AS2015. The trajectory illustrates a saturation of theoretical alignment at $\lambda_{\mathcal{I}} \approx 0.1$, reaching maximum interpretability before performance degrades.

Table 6. Pareto Sweep results for AS2009.

$\lambda_{\mathcal{I}}$	AUC	Alignment ($\bar{\mathcal{I}}$)	Interpretation
0.0 (Baseline)	0.8207	0.0293	Black Box
0.1 (Synergy)	0.8255	0.3846	Accuracy-Interpretability Synergy
0.2	0.8110	0.4270	Theory-Balanced Bias
0.3 (Saturation)	0.7963	0.4790	Theoretical Saturation
0.5	0.7760	0.4125	Grounding Degradation
0.8	0.7552	0.0708	Grounding Collapse
1.0	0.7501	0.0714	Grounding Collapse

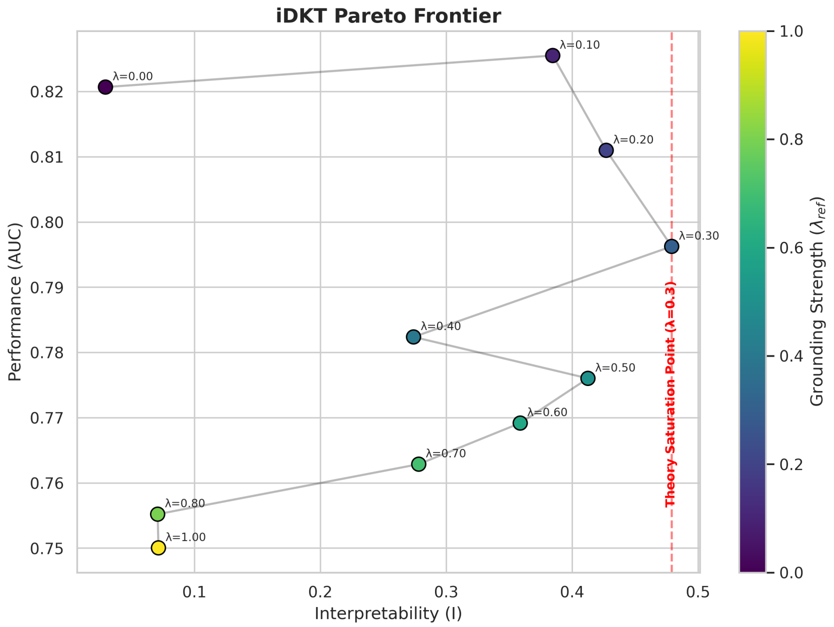


Figure 4. iDKT Pareto Frontier for AS2009. The curve displays an accuracy peak at $\lambda_{\mathcal{I}} = 0.1$, followed by a non-monotonic transition toward saturation and eventual collapse as constraints overwhelm behavioral patterns.

5.4. Improvement of Pedagogical Diagnostics (RQ3)

In the following sections, we analyze how the architectural design of iDKT can be leveraged to improve pedagogical diagnostics

5.4.1. Contextualization

The capability of iDKT to leverage longitudinal context is illustrated in Figure 5, which displays a "Twin Divergence" for a specific skill (Skill 4: Area Circle) in the AS2009 dataset. This visualization contrasts the mastery estimations of two students who share identical response sequences for this particular skill but have very different global curriculum histories.

While a Markovian model like BKT is mathematically forced to assign identical mastery states to both students due to their identical local sequences, iDKT leverages its self-attention mechanism to provide high-resolution contextual diagnostics through two distinct phenomena:

- *Non-Markovian Initialization*: At the very first interaction ($t = 0$), iDKT already provides divergent estimations. Student 2208 is recognized as having a significantly higher foundational baseline for this concept than Student 167, despite both having the same immediate local history.
- *Velocity-Driven Cross-Over*: Despite starting from a lower baseline, Student 167 exhibits a steeper learning curve (higher v_s). After the first success, their mastery estimation increases rapidly, eventually overtaking Student 2208 at the "Cross-Over Point" (approximately $t = 1.5$).

This dual evidence demonstrates that iDKT effectively decouples *Initial Standing* from *Learning Velocity*. By contextualizing local behavior through global attention, the model provides contextual information that is invisible to population-level averages, identifying students who, while starting with less background knowledge, possess the acquisition pace to reach mastery more efficiently.

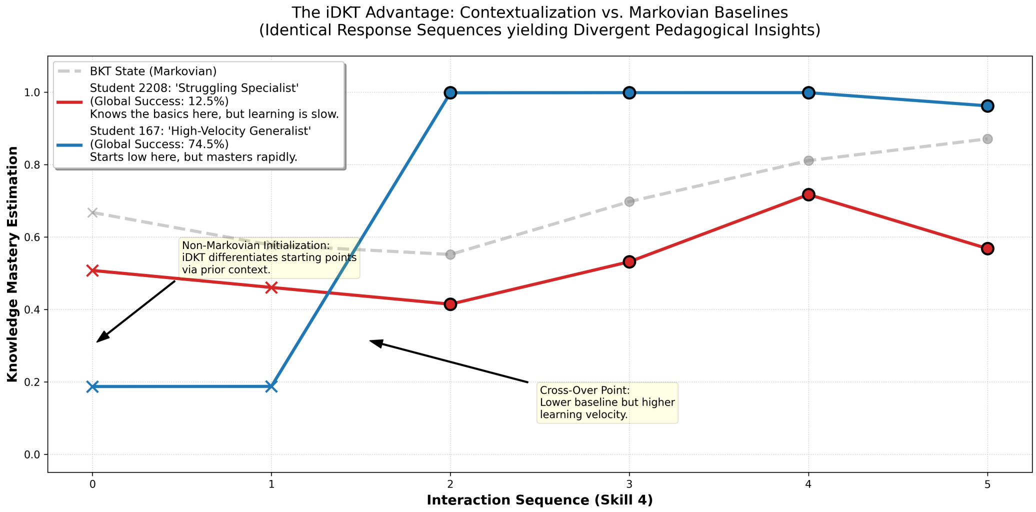


Figure 5. Twin Sequence Divergence for Skill 4 in AS2009. Two students with identical response sequences yield divergent pedagogical insights as iDKT contextualizes their local behavior within their global curriculum history, exhibiting both non-Markovian initialization and a learning velocity cross-over.

5.4.2. Individualization and Behavioral Heterogeneity

The primary contribution of iDKT, and the core of Hypothesis H3, lies in its ability to transform population-level theoretical averages into high-granularity contextual diagnostics. While standard BKT assumes that every student shares a fixed initial mastery (L_0) and learning rate (T) for a given skill, iDKT decomposes these into individualized profiles that capture personal learning velocities.

As illustrated in Figure 6, iDKT maps the student population into a continuous *diagnostic manifold* defined by two student-specific parameters: the Knowledge Gap (k_c, representing placement) and the Learning Velocity (v_s, representing pacing). Whereas a population-level model like BKT represents wait-time and acquisition as a single point—the theoretical mean—iDKT discovers a distributed landscape of student behaviors. Through unsupervised clustering, we identify four distinct learning archetypes: **Initial Standing** (Low k_c, Low v_s), **Developing Pacing** (Low k_c, Moderate v_s), **High-Velocity Growth** (Low k_c, High v_s), and **Advanced Standing** (High k_c, High v_s).

To statistically quantify this granularity, we analyze the *delta distribution* ($\Delta = p_{rate} - P(T)$), representing the student-specific deviation from the theoretical learning rate across all skills. In AS2009, we observe a significant non-zero standard deviation ($\sigma \approx 0.019$). This represents a fundamental increase in diagnostic resolution compared to the baseline BKT, where $\sigma = 0$ by definition. The presence of this variance confirms H3, proving that iDKT effectively captures the behavioral heterogeneity that classical models treat as a uniform average.

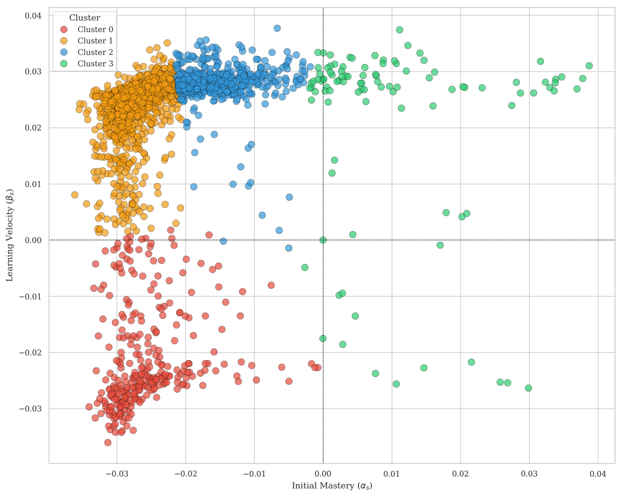


Figure 6. Discovery of student archetypes in the iDKT diagnostic space. Students are mapped along individualized Knowledge Gap (k_c) and Learning Velocity (v_s) axes, revealing the behavioral heterogeneity hidden within population-level BKT averages.

5.4.3. High-Granularity Trajectory Analysis

To provide visual evidence for hypothesis H3, we generated the mosaic showed in Figure 7 displaying individualized trajectories for three representative individual students (Fast, Median, Slow) per skill across the 15 most frequent skills in the dataset (including major topics like Circle Graphs, Venn Diagrams, and Linear Equations).

The qualitative analysis of these trajectories reveals three significant pedagogical insights:

1. **Dynamic Diagnostic Placement**: Even when BKT is forced to start at the population prior (L_0), iDKT starting points ($t = 1$) vary per student. This demonstrates that iDKT leverages cross-skill behavioral transfers to perform individualized placement before the first interaction with a new skill.
2. **Heterogeneous Learning Velocities**: We observe distinct slope gradients across students, confirming iDKT's ability to model individualized acquisition rates (t_s). While

Figure 7. Mastery Trajectory Alignment Mosaic for AS2009. Solid lines represent iDKT individualized mastery, while dashed lines show the BKT baseline. Markers indicate observed student performance (Circle: Correct, X: Incorrect). Each plot displays three unique students selected as archetypical representatives of Fast, Median, and Slow learning behaviors (non-aggregated).

BKT applies a fixed learning probability to all students, iDKT identifies that some learners acquire mastery rapidly (steep curves) while others progress more gradually (flatter curves), capturing the full spectrum of behavioral heterogeneity.

5.4.4. Archetype-Level Diagnostics

Individualized student profiling offers new opportunities for pedagogical monitoring and adaptive intervention. By projecting high-dimensional latent states into an interpretable skill-by-cohort matrix, iDKT provides educators with a comprehensive "knowledge blueprint" that supports data-driven differentiation.

This utility is visualized in Figure 8, which displays the mean latent mastery predicted by iDKT for four student clusters across 30 curriculum skills. The map reveals distinct "knowledge ceilings" and "learning frontiers" for each group, proving that the granularity discovered by the model translates into actionable insights for targeted instructional design. This result provides the final validation for H3, demonstrating that superior diagnostic granularity enables precision instruction.

5.4.5. Pedagogical Confidence

A significant practical application of this interpretability is evaluating the reliability of model predictions. By comparing BKT mastery estimations with those from iDKT, we can derive a measure of pedagogical confidence in the theoretical baseline. This relationship is visualized in Figure 9, which displays the concordance between BKT and iDKT mastery predictions across the most frequent skills and students.

The AS2015 results demonstrate high levels of agreement, whereas the AS2009 data reveals more divergence, where iDKT uncovers individualized patterns that deviate from the theoretical prior. These divergences serve as critical diagnostic indicators, identifying students or curriculum segments where the population-level model may be insufficient and personalized intervention is advisable. This result provides further validation for H3, demonstrating that superior diagnostic granularity directly translates into actionable pedagogical insights.

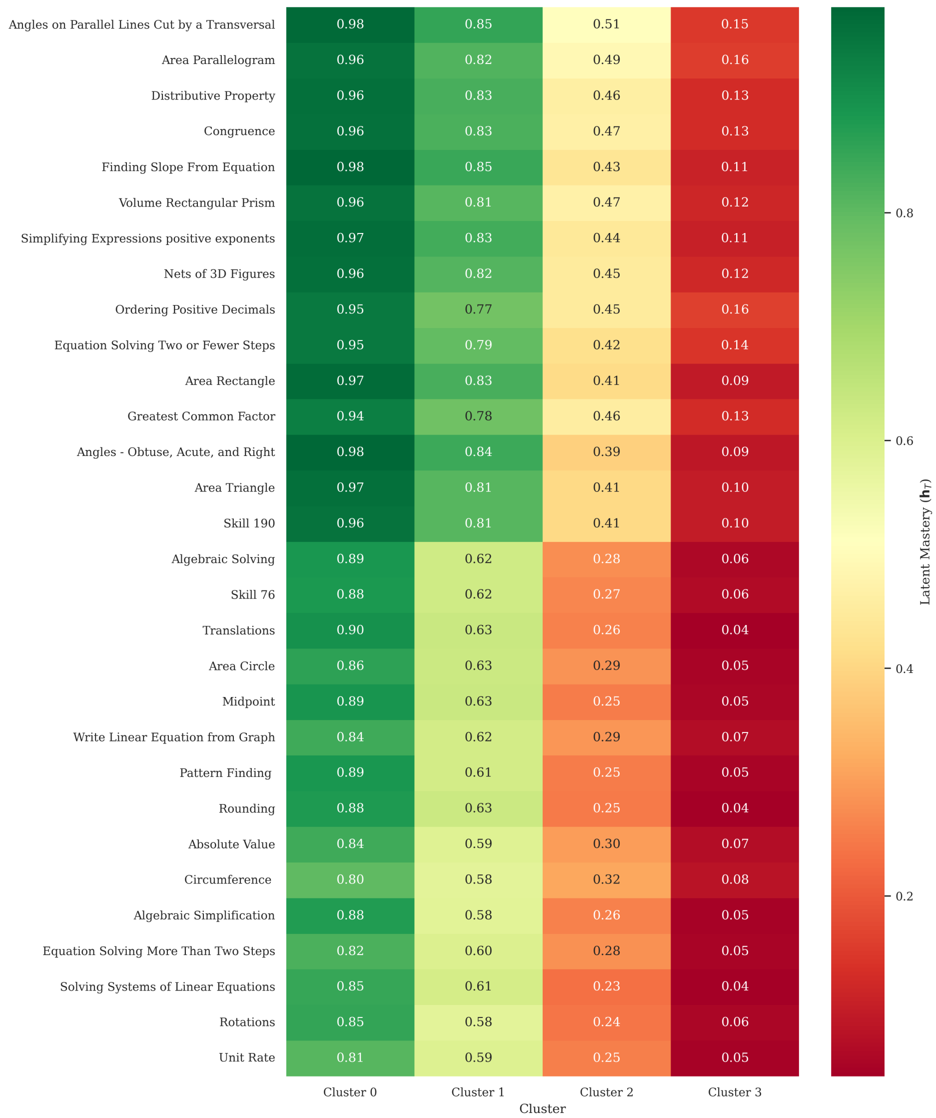
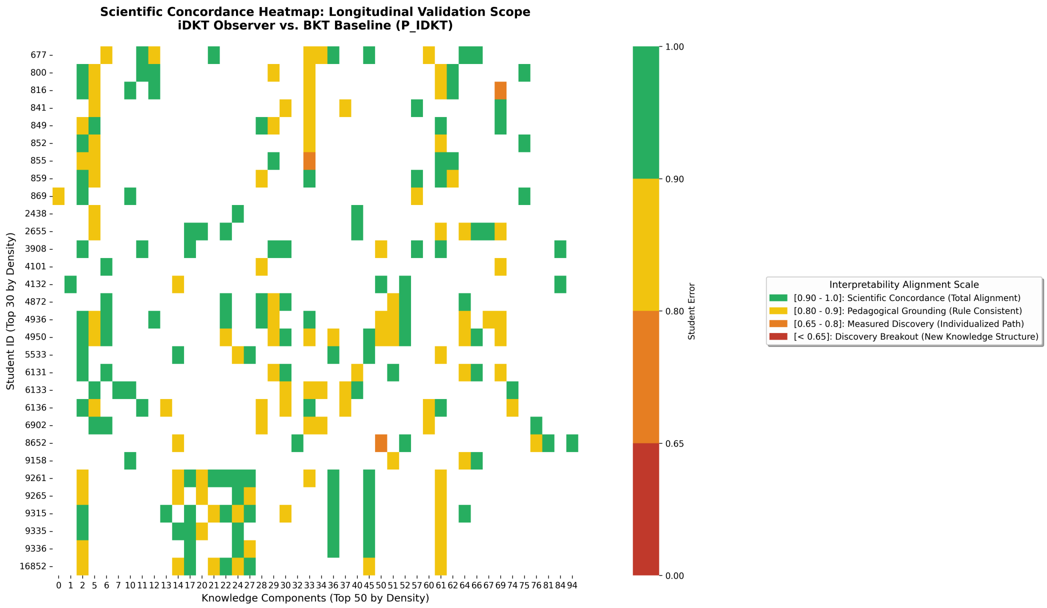
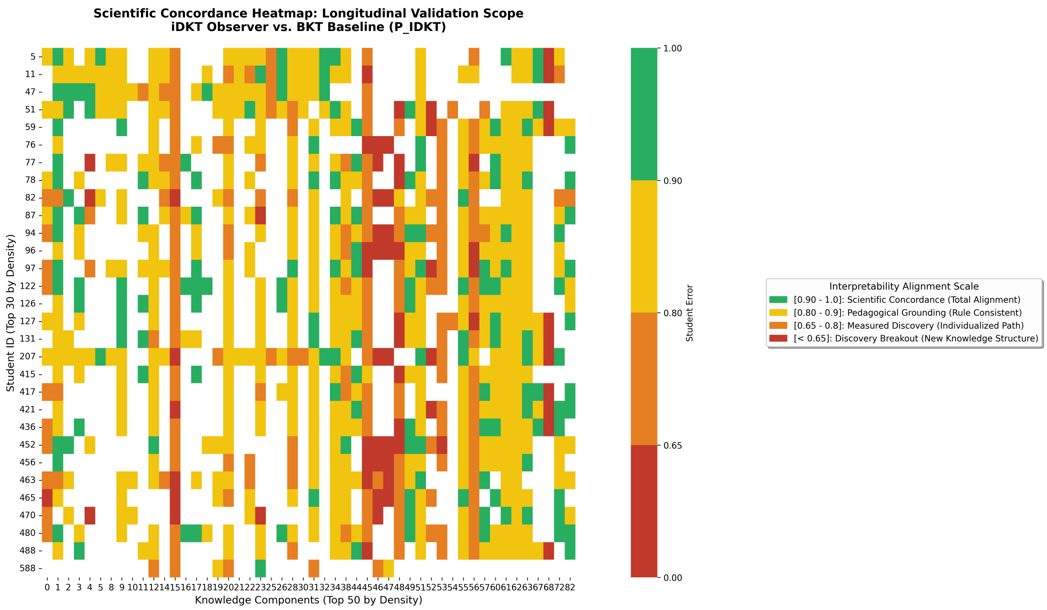


Figure 8. Pedagogical Diagnostic Map for AS2009. The heatmap displays the mean latent mastery predicted by iDKT for the four student archetypes across 30 curriculum skills. This visualization demonstrates the practical utility of iDKT's high-granularity estimations for targeted instructional design.



(a) AS2015



(b) AS2009

Figure 9. Heatmaps of per-skill prediction alignment.

6. Conclusions

This work has introduced iDKT, a Transformer-based Knowledge Tracing model that bridges the gap between the high predictive capacity of deep learning and the intrinsic interpretability of specialized pedagogical models. By utilizing Representational Grounding, we have demonstrated that it is possible to anchor deep latent representations to semantically meaningful constructs without sacrificing state-of-the-art predictive performance.

Our experimental results successfully address the proposed research questions. We demonstrated that iDKT successfully internalizes the theoretical constructs of the reference model, achieving high convergent validity ($\mathcal{I}_1$) and Robust Structural Encoding validated through diagnostic probing (selectivity $\Delta R^2 > 0.65$). This dual validation establishes that the semantic alignment reflects a genuine internal representation of knowledge states rather than superficial pattern matching. Furthermore, we formalized a methodology for exploring the accuracy–interpretability trade-off, identifying synergistic points where significant alignment is achieved with minimal impact on predictive AUC. Finally, we proved that iDKT provides a substantial increase in diagnostic granularity compared to population-level baselines, identifying individualized learning velocities that enable truly adaptive educational pacing.

The contribution of this research is twofold: a formal validation framework for evaluating the internal interpretability of deep learning models in education, and a practical architecture that transforms complex predictors into actionable pedagogical tools. By grounding deep learning in established domain concepts, iDKT offers a path toward AI-driven personalization that is both highly accurate and pedagogically interpretable, providing a robust foundation for personalization in intelligent tutoring systems.

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